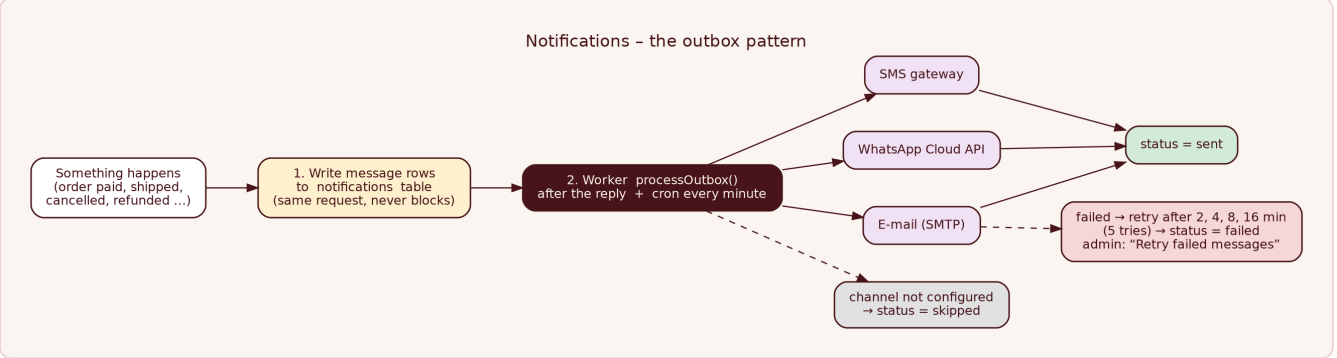
LUXE & THREAD · DOCUMENTATION

INTERNAL – for developers

Internal 12 · Notifications & Customer Marketing Design

Audience	INTERNAL – developers
Covers	E-mail, SMS, WhatsApp (optional), consent, campaigns, retries



1 · Design in one paragraph

Messages are first **written to a table** (`notifications`), then a worker sends them. A slow SMTP server can never slow or break checkout; failures retry automatically; nothing is lost. Each channel is optional: if it is not configured the message becomes `skipped`, not an error.

2 · Which events send what

Event	Template	Channels
Order placed & paid / COD confirmed	`order_confirmed`	e-mail · SMS* · WhatsApp**
Payment failed	`payment_failed` *(ready; wire when needed)*	e-mail
Handed to courier (SHIPPED)	`order_shipped` (courier, tracking link)	e-mail · SMS* · WhatsApp**
Delivered	`order_delivered`	same
Cancelled	`order_cancelled` (+ refund note)	same
Refund processed	`refund_processed`	same
Return approved / rejected	`return_update`	same
Offers / new arrivals	`marketing`	only the channel the customer **opted in** to

* if `SMS_ENABLED` · ** if `WHATSAPP_ENABLED` **and** the customer ticked WhatsApp.

3 · Channels, providers, India rules

Channel	How it is sent	Open-source / alternatives	India-specific rules
E-mail	Nodemailer → any SMTP	Postfix, Mailu, Mailcow (self-host); or provider SMTP (Amazon SES, Brevo, Zoho)	SPF + DKIM + DMARC on your domain; unsubscribe link in marketing
SMS	HTTP POST to your gateway (`SMS_GATEWAY_URL`) JSON `{to,message,sender}`	gateways: MSG91, Fast2SMS, Kaleyra… (self-host Kannel/Gammu needs a GSM modem + operator DLT)	**TRAI DLT registration** of your *entity, sender ID (header) and each message template* is mandatory; the text must match the approved template; promotional SMS need consent and timing rules. Adjust `sendSms` payload to your gateway's API
WhatsApp	Meta WhatsApp Business **Cloud API**, template messages	Official API is free to set up, conversation pricing applies; open-source gateways (Evolution API, WAHA) are **unofficial** and risk number bans – not recommended for a business	Templates must be **pre-approved** by Meta; customer must opt in; template names = our names (`order_confirmed` …) with body variables `{{1}}` first name, `{{2}}` order id

Suggested approved texts (copy to your DLT / Meta console and adjust)

- `order_confirmed`: "Hi {{1}}, your Luxe & Thread order {{2}} is confirmed. Track it: {{3}}"
- `order_shipped`: "Hi {{1}}, order {{2}} is on its way via {{3}}. Tracking: {{4}}"
- `order_cancelled`: "Hi {{1}}, order {{2}} is cancelled. Any payment will be refunded in 5–7 days." (The code sends `params: [first name, order id]` for WhatsApp; extend `notifyOrder` if you add more variables.)

4 · Consent (DPDP Act 2023 / TRAI)

- Checkout has **three separate, un-ticked** boxes (e-mail, SMS, WhatsApp). Order updates are *transactional* and always sent by e-mail.
- Consent flags are only ever switched **on** by an explicit tick; **off** by the unsubscribe link (`/unsubscribe?t=<token>` → confirm button → all marketing consent off).
- `consent_updated_at` records when consent last changed. Keep the Privacy Policy accurate.
- Marketing never goes to customers without the flag; `queueCampaign` filters `*_opt_in = true`.

5 · Campaign procedure (owner)

Admin → Customers → *Send offers* → name, channel, subject, message → **Queue campaign**. The counts of eligible customers are shown first. Messages go out within ~1 minute (cron), personalised with "Hi ,". Check Admin → Messages.

6 · Worker details

`processOutbox(limit)`: `UPDATE … WHERE id IN (SELECT … FOR UPDATE SKIP LOCKED) RETURNING` claims rows (attempts+1, `next_attempt_at = now+10 min` as a lease) → send → `sent` | `skipped` | on error `markFailed(retry in 2^attempts min)`; after 5 attempts `failed`. Triggered by `after()` on order events and by cron `/api/jobs/notifications` every minute.

7 · Operating & monitoring

`/api/health` → `notificationQueue` counts; Admin → Messages (last error per row); alert if `failed > 0` or `queued` grows. Purge old rows monthly (Database doc §8).

8 · Extending

New event → add a template in `templates.ts`, call `notifyOrder(order, 'name')` at the service step, add a unit test. New channel → add a provider in `providers.ts` and a case in `deliver()`.